

Product Spin(7)-instantons on $X \times X$ and the linear problem

Notes on gluing pulled-back ASD instantons at the type (ii) and (iii) singularities of Joyce's first Spin(7)-manifold (companion to arXiv:2310.03451)

prepared for Daniel Platt

August 29, 2026

Abstract

In arXiv:2310.03451 (Galdeano–Platt–Tanaka–Wang, henceforth [GPTW]) non-flat ASD instantons are glued in only along the type (i) singularities $S_{73}, \dots, S_{76}$ of T^8/Γ ; along the type (ii) singularities $S_1, \dots, S_8$ and at the type (iii) points $S_9, \dots, S_{72}$ (resolved by $X \times X$, $X = \text{Eguchi–Hanson}$) the flat connection and the trivial connection are used (Definition 4.6 (iv)). These notes work out what is needed to glue pulled-back ASD instantons $p_1^* A^\alpha$ along $\text{fix}(\alpha)$ and $p_2^* A^\beta$ along $\text{fix}(\beta)$ as well. (1) The *product connection* $B = p_1^* A^\alpha \boxtimes p_2^* A^\beta$ on $X \times X$ is a Spin(7)-instanton (in fact Hermitian Yang–Mills for every product Kähler structure) on a bundle whose restriction to the two ends of $X \times X$ is exactly the pulled-back bundle appearing on the neighbouring type (ii) pieces (Section 2). (2) The analogue of [GPTW, Section 5.2] is proved: if A^α and A^β are infinitesimally rigid as G -connections, then $\text{Ker } L_B = \text{Ker } L_B^* = 0$ on $C_\beta^{1,\alpha}$ for every $\beta < 0$ (Theorem 4.1), and the corresponding statement on the flat cone holds (Proposition 4.2). The proof reduces, via the spinor description of the Spin(7) deformation operator on a product of two hyperkähler 4-manifolds, to [GPTW, Prop. 5.8] plus a separation-of-variables argument; without the rigidity hypothesis the cokernel is computed explicitly in terms of L^2 and bounded harmonic spinors on X , and the kernel is identified for the constituents of $\text{Ad } Q$ pulled back from one factor. (3) Since rigidity is now needed for all representations of the structure group appearing in $\mathfrak{g}$, Section 6 determines which $U(1)$ -instantons L^t on X are rigid in this sense: the L^2 -kernel of the Dirac operator on $S^- \otimes L^t$ has dimension $\lfloor t^2/4 \rfloor$. This has a consequence for [GPTW, Prop. 7.8] that the authors may want to double check (Remark 6.2). An explicit $SO(5)$ example of compatible gluing data with $\text{hol}_\theta(\alpha) \neq 1 \neq \text{hol}_\theta(\beta)$ is given (Proposition 6.3); the linear-algebra and representation-theoretic checks were done by computer (scripts attached).

Contents

1	Set-up and notation	2
2	The product connection	3
2.1	Definition and first properties	3
2.2	Compatible gluing data with instantons at the type (ii)/(iii) singularities	4
3	Spinor description of the linearised operator	5
3.1	Dirac operators on the factors and on the product	5
4	The analogue of Section 5.2	7
5	Consequences for Section 6 and the gluing theorem	10
6	Which instantons on X are G-rigid? The $U(1)$ case	10

1 Set-up and notation

Throughout we use the notation of [GPTW]. X is the Eguchi–Hanson space with blow-up map $\rho: X \rightarrow \mathbb{C}^2/\{\pm 1\}$, $r := |\rho|$, hyperkähler triple $(\omega_1, \omega_2, \omega_3)$, oriented so that the ω_a are self-dual; ASD instantons have curvature in Λ_-^2 . $E \subset X$ denotes the exceptional sphere. On $X \times X$ we write p_1, p_2 for the two projections, $r_1 = r \circ p_1$, $r_2 = r \circ p_2$, and

$$w(x, y) = 1 + \min\{r(x), r(y)\}$$

for the weight function of [GPTW, Def. 5.7]; $C_\beta^{k, \alpha}$ denotes the weighted Hölder spaces defined with this weight.

The product Spin(7)-structure. By [GPTW, Lemma 5.2] the four-form

$$\Omega = \frac{1}{2} \omega_1 \wedge \omega_1 + \frac{1}{2} \mu_1 \wedge \mu_1 - \sum_{a=1}^3 \omega_a \wedge \mu_a \quad (1)$$

is a torsion-free Spin(7)-structure on $X \times X$, where (ω_a) is the triple on the first factor and (μ_a) a suitably oriented hyperkähler triple on the second factor (in the coordinates of [GPTW, §2.1], with $\omega = (dx_{12} + dx_{34}, dx_{13} - dx_{24}, dx_{14} + dx_{23})$ one may take $\mu = (-dx_{56} - dx_{78}, -dx_{57} + dx_{68}, dx_{58} + dx_{67})$, and then (1) is exactly Ω_0). The following decomposition is elementary; I checked it numerically with the script `check_spin7.py` (eigenvalues of $\alpha \mapsto *(\Omega_0 \wedge \alpha)$ and projections of the relevant forms).

Lemma 1.1. *With respect to Ω one has*

$$\begin{aligned} \Lambda_7^2 &= \langle \omega_a - \mu_a : a = 1, 2, 3 \rangle \oplus \Lambda_{7, \text{mix}}^2, & \Lambda_{7, \text{mix}}^2 &\subset p_1^* \Lambda^1 \otimes p_2^* \Lambda^1, \quad \dim \Lambda_{7, \text{mix}}^2 = 4, \\ \Lambda_{21}^2 &\supset p_1^* \Lambda_-^2 \oplus p_2^* \Lambda_-^2 \oplus \langle \omega_a + \mu_a \rangle. \end{aligned}$$

In particular $p_1^ \Lambda_-^2(X) \oplus p_2^* \Lambda_-^2(X) \subset \Lambda_{21}^2$, and π_7^2 restricted to $p_i^* \Lambda_+^2$ is injective (with $|\pi_7^2 \omega_a| = |\omega_a|/\sqrt{2}$).*

The three parallel sections $\omega_a - \mu_a$ of Λ_7^2 are the Kähler forms of the three product complex structures $I_a \times (-J_a)$; for each of them $X \times X$ is a (non-compact) Calabi–Yau four-fold and Ω is the associated Cayley form.

Spinors. X is spin (it is simply connected) and hyperkähler, so with the orientation above $S^+(X)$ is trivialised by parallel spinors while $S^-(X)$ has none. Clifford multiplication by Λ_-^2 is trivial on S^+ and by Λ_+^2 is trivial on S^- . On the flat end $(\mathbb{C}^2 \setminus B)/\{\pm 1\}$ the deck involution lifts to the spin structure of X as the element of Spin(4) that acts by +1 on S^+ (this is forced by the existence of parallel spinors) and hence by −1 on S^- ; we record this as

$$-1 \in \mathbb{Z}_2 \text{ acts on constant spinors of the end by } +1 \text{ on } S^+(\mathbb{R}^4), \quad -1 \text{ on } S^-(\mathbb{R}^4). \quad (2)$$

For a bundle $\mathcal{V} \rightarrow X$ with connection asymptotic (at rate −3, [GPTW, Prop. A.5]) to the flat connection with holonomy $g \in \text{Aut}(V)$, $g^2 = 1$, the constant sections of $S^\pm \otimes \mathcal{V}$ on the end are therefore $S^\pm(\mathbb{R}^4) \otimes V^\pm$, where $V^\pm$ is the (± 1) -eigenspace of g .

On the product, $S(X \times X) = S(X_1) \hat{\otimes} S(X_2)$ and

$$S^+(X \times X) = S_1^+ \otimes S_2^+ \oplus S_1^- \otimes S_2^-, \quad S^-(X \times X) = S_1^- \otimes S_2^+ \oplus S_1^+ \otimes S_2^-. \quad (3)$$

The parallel spinors of $X \times X$ form the four-dimensional space $S_1^+ \otimes S_2^+$ (holonomy $SU(2) \times SU(2)$); the $\text{Spin}(7)$ -structure (1) corresponds to a unit vector $\psi \in S_1^+ \otimes S_2^+$ and, as on any $\text{Spin}(7)$ -manifold, Clifford multiplication on ψ gives parallel isomorphisms

$$\Lambda^1 \xrightarrow{\cong} S^-, \quad v \mapsto v \cdot \psi, \quad \Lambda^0 \oplus \Lambda_7^2 \xrightarrow{\cong} S^+, \quad (f, \eta) \mapsto f\psi + \eta \cdot \psi. \quad (4)$$

Under (4) and (3): $p_1^* \Lambda^1 \leftrightarrow S_1^- \otimes S_2^+$, $p_2^* \Lambda^1 \leftrightarrow S_1^+ \otimes S_2^-$, $\Lambda^0 \oplus \langle \omega_a - \mu_a \rangle \leftrightarrow S_1^+ \otimes S_2^+$, and $\Lambda_{7,\text{mix}}^2 \leftrightarrow S_1^- \otimes S_2^-$.

2 The product connection

2.1 Definition and first properties

Let G be a compact Lie group and let $H_1, H_2 \subset G$ be closed subgroups which commute element-wise, so that $\iota: H_1 \times H_2 \rightarrow G$, $(h_1, h_2) \mapsto h_1 h_2$, is a homomorphism. Let (P_1, A_1) be a principal H_1 -bundle over X with a finite-energy ASD instanton asymptotic to the flat connection with holonomy $g_1 \in H_1$ at infinity, and (P_2, A_2) a principal H_2 -bundle with finite-energy ASD instanton asymptotic to the flat connection with holonomy $g_2 \in H_2$. Put

$$Q := (p_1^* P_1 \times_{X \times X} p_2^* P_2) \times_{H_1 \times H_2} G \longrightarrow X \times X, \quad B := \text{the connection induced by } (p_1^* A_1, p_2^* A_2), \quad (5)$$

We call (Q, B) the *product connection*. Its adjoint bundle is

$$\text{Ad } Q = (p_1^* P_1 \times p_2^* P_2) \times_{H_1 \times H_2} \mathfrak{g} = \bigoplus_k p_1^* \mathcal{V}_k \otimes p_2^* \mathcal{W}_k =: \bigoplus_k \mathcal{V}_k \boxtimes \mathcal{W}_k, \quad (6)$$

where $\mathfrak{g}|_{H_1 \times H_2} = \bigoplus_k V_k \otimes W_k$ is the decomposition into irreducible $H_1 \times H_2$ -modules (with multiplicities), $\mathcal{V}_k = P_1 \times_{H_1} V_k$, $\mathcal{W}_k = P_2 \times_{H_2} W_k$, and the induced connection on $\mathcal{V}_k \boxtimes \mathcal{W}_k$ is $\nabla^{A_1} \otimes 1 + 1 \otimes \nabla^{A_2}$. Here $\mathfrak{g}$ is regarded as a real representation and “irreducible” may be taken over $\mathbb{R}$ or, after complexification, over $\mathbb{C}$; nothing below depends on this.

Proposition 2.1. (i) B is a $\text{Spin}(7)$ -instanton for Ω ; more precisely $F_B = \iota_*(p_1^* F_{A_1} + p_2^* F_{A_2}) \in \Omega_{21}^2(\text{Ad } Q)$, and B is Hermitian Yang–Mills for each of the three product Kähler structures.

(ii) $|\nabla^k F_B| \leq c((1+r_1)^{-4-k} + (1+r_2)^{-4-k})$; in particular $\|F_B\|_{C_{-2}^{0,\alpha}} < \infty$ for the weight w .

(iii) (Topology of the two ends.) Let θ_2 be the flat H_2 -connection on the end $\{r_2 > R\} \cong (\mathbb{C}^2 \setminus B_R)/\{\pm 1\}$ of the second factor with holonomy g_2 , and let Φ_2 be the framing of [GPTW, Def. A.3]. On $X \times \{r_2 > R\}$ the bundle Q is isomorphic to $(p_1^* P_1 \times p_2^* E_{\infty,2}) \times_{H_1 \times H_2} G$, and under this isomorphism

$$B - (p_1^* A_1, p_2^* \theta_2) = p_2^*(\Phi_2^* A_2 - \theta_2) = O(r_2^{-3}), \quad \nabla^k(\dots) = O(r_2^{-3-k}).$$

The same holds with the roles of the factors exchanged. In other words, on the end $\{r_2 > R\}$ the product connection is, up to an $O(r_2^{-3})$ error, the pull-back of A_1 from the first factor twisted by the flat connection with holonomy g_2 in the second factor – which is exactly the connection glued in along the type (ii) locus $\text{fix}(\alpha)$ (Section 2.2).

Proof. (i) The curvature of the product connection on $p_1^* P_1 \times p_2^* P_2$ is $p_1^* F_{A_1} + p_2^* F_{A_2}$ with values in $\mathfrak{h}_1 \oplus \mathfrak{h}_2$; $F_{A_i} \in \Omega_-^2(X)$, so by Lemma 1.1 the curvature lies in Λ_{21}^2 , and $\pi_7^2 F_B = 0$. For the Kähler structure $I_a \times (-J_a)$ with Kähler form $\omega_a - \mu_a$ the curvature is of type $(1,1)$ on each factor (ASD forms are $(1,1)$ for every complex structure of the hyperkähler family) and $\Lambda_{\omega_a - \mu_a} F_B = \Lambda_{\omega_a} F_{A_1} - \Lambda_{\mu_a} F_{A_2} = 0$ since ASD forms are primitive. (ii) follows from [GPTW, Prop. A.5] applied to each factor. (iii) is a restatement of the framing at infinity of (P_2, A_2) : on $\{r_2 > R\}$ we have $\Phi_2: E_{\infty,2} \cong P_2$ and $\Phi_2^* A_2 - \theta_2 = O(r_2^{-3})$; pulling back by p_2 and forming the fibre product with $p_1^* P_1$ gives the claim. $\square$

Example 2.2 (direct sums of $U(1)$ -instantons). Let $G = SO(n)$, $n \geq 4$, $H_1 = SO(2)$ acting on the coordinates 1, 2 and $H_2 = SO(2)$ acting on the coordinates 3, 4. Let (L, A) be the Gocho–Nakajima $U(1)$ -instanton of [GPTW, Prop. A.7] ($c_1(L)$ the generator of $H^2(X; \mathbb{Z}) \cong \mathbb{Z}$, holonomy -1 at infinity) and L^t its tensor powers. Then

$$E^\alpha = [L^{t_1}] \oplus \mathbb{R}^{n-2}, \quad E^\beta = \mathbb{R}^2 \oplus [L^{t_2}] \oplus \mathbb{R}^{n-4}, \quad Q = p_1^*[L^{t_1}] \oplus p_2^*[L^{t_2}] \oplus \mathbb{R}^{n-4}$$

with the induced connections, and $g_1 = \text{diag}(-1, -1, 1, \dots)$, $g_2 = \text{diag}(1, 1, -1, -1, 1, \dots)$ if t_1, t_2 are odd. The adjoint bundle is

$$\text{Ad } Q = \Lambda^2 Q = \mathbb{R}^2 \oplus \Lambda^2 \mathbb{R}^{n-4} \oplus (n-4) p_1^*[L^{t_1}] \oplus (n-4) p_2^*[L^{t_2}] \oplus [L^{t_1}] \boxtimes [L^{t_2}],$$

so exactly one “doubly twisted” summand $[L^{t_1}] \boxtimes [L^{t_2}]$ occurs. Tensor-product constructions ($SU(2) \times SU(2) \rightarrow SO(4)$, $U(n_1) \times U(n_2) \rightarrow U(n_1 n_2)$, ...) are covered by the same formalism.

2.2 Compatible gluing data with instantons at the type (ii)/(iii) singularities

Recall from [GPTW, §2.2] that near a type (iii) point the orbifold is $B_\zeta^4/\{\pm 1\} \times B_\zeta^4/\{\pm 1\}$ with coordinates (x, y) , $x = (x_1, \dots, x_4)$, $y = (x_5, \dots, x_8)$, $\alpha(x, y) = (-x, y)$, $\beta(x, y) = (x, -y)$; $\text{fix}(\alpha) = \{x = 0\}$ is resolved by $X_x \times (T^4/\{\pm 1\})_y$, $\text{fix}(\beta) = \{y = 0\}$ by $(T^4/\{\pm 1\})_x \times X_y$, and the point by $X_x \times X_y$. The generalisation of [GPTW, Def. 4.6] is:

Definition 2.3. *Extended compatible gluing data* consist of

- (i) a flat orbifold G -bundle $(E_0, \theta) \rightarrow T^8/\Gamma$, i.e. a representation $\varrho: \pi_1^{\text{orb}}(T^8/\Gamma) \rightarrow G$, which is infinitesimally rigid, irreducible and unobstructed as a $\text{Spin}(7)$ -instanton (Lemma 7.1 of [GPTW]); write $g_\alpha = \varrho(\alpha)$, $g_\beta = \varrho(\beta)$ (commuting involutions);
- (ii) framings ϕ_k as in [GPTW, Def. 4.6 (ii)];
- (iii) for $j \in \{73, \dots, 76\}$ bundles (E_j, A_j) as in [GPTW, Def. 4.6 (iii)];
- (iv) for the components of $\text{fix}(\alpha)$ a G -bundle $(E^\alpha, A^\alpha) \rightarrow X$ with a finite-energy ASD instanton, infinitesimally rigid as a G -connection, asymptotic to the flat connection with holonomy g_α , such that $\varrho(\beta), \varrho(\tau_5), \dots, \varrho(\tau_8)$ act on (E^α, A^α) by automorphisms compatible with the framing at infinity; similarly (E^β, A^β) for $\text{fix}(\beta)$ with holonomy g_β , on which $\varrho(\alpha), \varrho(\tau_1), \dots, \varrho(\tau_4)$ act by automorphisms;
- (v) the structure groups of E^α, E^β reduce to commuting subgroups $H_1, H_2 \subset G$ (i.e. $E^\alpha = P_1 \times_{H_1} G$, $E^\beta = P_2 \times_{H_2} G$ with A^α, A^β induced from H_i -connections A_i), so that at every type (iii) point the product connection (5) is defined.

Condition (iv) replaces (iv) of [GPTW, Def. 4.6]: the monodromies of θ around $S_1, \dots, S_8$ are no longer required to be trivial but to agree with the holonomies at infinity of A^α, A^β ; the monodromy around a type (iii) point is then $g_\alpha g_\beta$, which is the holonomy at infinity of B along the diagonal directions, as it must be. The automorphism condition in (iv) is needed to form the bundle $(p_1^* E^\alpha)/(\mathbb{Z}^4 \rtimes \mathbb{Z}_2)$ over $X_x \times (T^4/\{\pm 1\})_y$, exactly as in the definition of $\tilde{E}_j$ for $j \in \{1, \dots, 8\}$ in [GPTW, (4.12)]; in Example 2.2 it means that these group elements act by ± 1 on the $[L^t]$ -summand.

Bundle and approximate solution. Define $\tilde{E}_j$ for $j \in \{1, \dots, 8\}$ as $(p_1^* E^\alpha \rightarrow X_x \times \mathbb{R}_y^4)/(\mathbb{Z}^4 \rtimes \mathbb{Z}_2)$ (resp. with E^β), for $j \in \{9, \dots, 72\}$ as Q , and for $j \in \{73, \dots, 76\}$ as in [GPTW]. By Proposition 2.1 (iii) the identifications on overlaps exist and E_t is defined by

[GPTW, (4.12)]. The approximate solution is obtained by cutting off *each factor separately*: on the α -chart $X_x \times (T^4/\{\pm 1\})_y$ put

$$A_t = (\theta_x + \chi(x)a^\alpha(x)) \boxtimes (\theta_y + \chi(y)a^\beta(y)), \quad a^\alpha = p_1^* A^\alpha - \theta_x, \quad a^\beta = p_2^* A^\beta - \theta_y,$$

with χ the cut-off of [GPTW, (4.13)] in the respective variable (so that near a type (iii) point $A_t = B$, in the annuli $\zeta/4 \leq |t\rho(x)| \leq \zeta/2$ resp. $\zeta/4 \leq |t\rho(y)| \leq \zeta/2$ one factor is being interpolated to the flat connection, and away from all annuli $A_t = \theta$). Since the connection is a product, $F_{A_t} = F^{(1)} \otimes 1 + 1 \otimes F^{(2)}$ where $F^{(i)}$ is the curvature of the cut-off connection in the i -th factor, and [GPTW, (4.19)–(4.21)] apply verbatim to each factor. Proposition 2.1 (ii) gives $\|F_{A_t}\|_{C_{-2;t}^{0,\alpha}} \leq c$ on the $X \times X$ pieces (use $r_t = t \min\{r_1, r_2\}$ there). Hence:

Proposition 2.4 (pregluing estimate). *For extended compatible gluing data the approximate solution satisfies $\|(\pi_7^2)^{\Omega_t} F_{A_t}\|_{C_{-2;t}^{0,\alpha}} \leq ct^{3/10}$, with the proof of [GPTW, Prop. 4.8] unchanged.*

Remark 2.5. As in [GPTW, Remark 4.7], $p_1(\text{Ad } E_t)$ acquires additional terms $-k^\alpha \text{P.D.}[\hat{S}_\alpha] - k^\beta \text{P.D.}[\hat{S}_\beta]$ with $k^\alpha = \frac{1}{8\pi^2} \int_X |F_{A^\alpha}|^2$, where $\hat{S}_\alpha \in H_4(M_t; \mathbb{Z})$ is the class of the proper transform of $\{x_0\} \times T_y^4/\{\pm 1\}$ (a Kummer-type $K3$ surface after the type (iii) resolutions); different instanton numbers give non-gauge-equivalent instantons.

3 Spinor description of the linearised operator

3.1 Dirac operators on the factors and on the product

For a Hermitian bundle $(\mathcal{V}, \nabla)$ over X with ASD connection write $D_{\mathcal{V}} = D_{\mathcal{V}}^+ \oplus D_{\mathcal{V}}^-$ for the twisted Dirac operator, $D_{\mathcal{V}}^\pm: \Gamma(S^\pm \otimes \mathcal{V}) \rightarrow \Gamma(S^\mp \otimes \mathcal{V})$. The Lichnerowicz formula and the fact that Λ_-^2 acts trivially on S^+ give

$$D_{\mathcal{V}}^- D_{\mathcal{V}}^+ = \nabla^* \nabla \quad \text{on } S^+ \otimes \mathcal{V}, \quad D_{\mathcal{V}}^+ D_{\mathcal{V}}^- = \nabla^* \nabla + \mathcal{F}_{\mathcal{V}} \quad \text{on } S^- \otimes \mathcal{V}, \quad (7)$$

where $\mathcal{F}_{\mathcal{V}} = \sum_{i < j} c(e_i e_j) \otimes F_{\nabla}(e_i, e_j)$ is the curvature term (there is no scalar curvature term). Since S^+ is trivialised by parallel spinors, $\Lambda^1 \otimes \mathcal{V} \otimes \mathbb{C} = S^+ \otimes (S^- \otimes \mathcal{V})$ and $(\Lambda^0 \oplus \Lambda_+^2) \otimes \mathcal{V} \otimes \mathbb{C} = S^+ \otimes (S^+ \otimes \mathcal{V})$, and the ASD deformation operator $\tilde{\delta}_{\nabla} = (d_{\nabla}^*, \sqrt{2} d_{\nabla}^+)$ of [GPTW, proof of Prop. 7.8] is $\text{id}_{S^+} \otimes D_{\mathcal{V}}^-$. Consequently

$$\text{Ker}_{L^2} \delta_{\nabla} \otimes \mathbb{C} \cong S^+ \otimes \text{Ker}_{L^2} D_{\mathcal{V}}^-, \quad \text{so} \quad \text{Ker}_{L^2} \delta_{\nabla} = 0 \iff \text{Ker}_{L^2} D_{\mathcal{V}}^- = 0. \quad (8)$$

Definition 3.1. For $(\mathcal{V}, \nabla)$ as above put

$$\begin{aligned} \mathcal{K}(\mathcal{V}) &:= \{\varphi \in \Gamma(S^- \otimes \mathcal{V}) : D^- \varphi = 0, \varphi \in L^2\}, \\ \mathcal{K}^b(\mathcal{V}) &:= \{\varphi \in \Gamma(S^- \otimes \mathcal{V}) : D^- \varphi = 0, \varphi \text{ bounded}\}, \\ \mathcal{K}^+(\mathcal{V}) &:= \{\varphi \in \Gamma(S^+ \otimes \mathcal{V}) : D^+ \varphi = 0, \varphi \in L^2\}, \\ \mathcal{K}^{+,b}(\mathcal{V}) &:= \{\varphi \in \Gamma(S^+ \otimes \mathcal{V}) : D^+ \varphi = 0, \varphi \text{ bounded}\}. \end{aligned}$$

Lemma 3.2 (harmonic spinors on X). *Let $(\mathcal{V}, \nabla)$ be asymptotic at rate -3 to the flat connection with holonomy g at infinity, and let φ be a bounded solution of $D^\pm \varphi = 0$.*

(i) *There is a constant spinor $\varphi_\infty \in S^\pm(\mathbb{R}^4) \otimes V^\pm$ (notation of (2)) such that, in the framing at infinity, $\varphi - \varphi_\infty = O(r^{-2})$; if $\varphi \rightarrow 0$ at infinity then $\varphi = O(r^{-3})$, in particular $\varphi \in L^2$. All spaces in Definition 3.1 are finite dimensional, $\mathcal{K} \subset \mathcal{K}^b$, and $\varphi \mapsto \varphi_\infty$ induces an injection $\mathcal{K}^b/\mathcal{K} \hookrightarrow S^-(\mathbb{R}^4) \otimes V^-$.*

(ii) $\mathcal{K}^+(\mathcal{V}) = 0$. *If $\varphi \in \mathcal{K}^{+,b}(\mathcal{V})$ and $\varphi \rightarrow 0$ at infinity then $\varphi = 0$.*

(iii) If g acts trivially on V (e.g. $\mathcal{V} = \mathbb{R}$ with the trivial connection), then $\mathcal{K}^b(\mathcal{V}) = \mathcal{K}(\mathcal{V})$. For the trivial bundle, $\mathcal{K}(\mathbb{R}) = \mathcal{K}^b(\mathbb{R}) = 0$ and $\mathcal{K}^{+,b}(\mathbb{R}) = S^+$ consists of the parallel spinors.

Proof. (i) On the end, $D^\pm = D_{\text{flat}}^\pm + P$ where P is a first-order operator with coefficients $O(r^{-3})$ (connection) and $O(r^{-4})$ (metric), cf. [GPTW, Def. A.1 and Prop. A.5]. The indicial roots of $D_{\text{flat}}^\pm$ on the cone $(\mathbb{R}^4 \setminus 0)/\{\pm 1\}$ are $\{0, 1, 2, \dots\} \cup \{-3, -4, \dots\}$ (harmonic spinors on $\mathbb{R}^4$ homogeneous of degree k are polynomial, of degree $-3 - k$ their Kelvin transforms; the eigenvalues of the Dirac operator on S^3 are $\pm(k + \frac{3}{2})$), and the degree-0 solutions that descend to the quotient are exactly the constants in $S^\pm(\mathbb{R}^4) \otimes V^\pm$. Standard weighted elliptic theory on the end (e.g. the argument of [Walpuski, Prop. 2.41] = [GPTW, Prop. A.6], or Lockhart–McOwen) gives: a solution bounded by r^ϵ is $\varphi_\infty + O(r^{-2})$ (the $O(r^{-2})$ term is produced by the source $P\varphi_\infty = O(r^{-3})$; -2 is not an indicial root), and a solution of $o(1)$ is $O(r^{-3})$. Finite dimensionality follows from the Fredholm property on weighted spaces with weight $\epsilon \in (0, 1)$; the last claim is the definition of φ_∞ . (ii) By (7) $\nabla^* \nabla \varphi = 0$; for $\varphi \in L^2$ (which by (i) holds as soon as $\varphi \rightarrow 0$) integration by parts gives $\nabla \varphi = 0$, and a parallel section which decays vanishes. (iii) If $g = 1$ on V then $V^- = 0$, so $\varphi_\infty = 0$ for $\varphi \in \mathcal{K}^b$ and (i) gives $\varphi \in L^2$. For $\mathbb{R}$ this shows $\mathcal{K}^b = \mathcal{K}$, and $\mathcal{K} = 0$ because $D^+ D^- = \nabla^* \nabla$ on S^- (untwisted) and decaying parallel spinors vanish. Finally a bounded $\varphi \in \mathcal{K}^{+,b}(\mathbb{R})$ satisfies $\nabla^* \nabla \varphi = 0$ with $\varphi = \varphi_\infty + O(r^{-2})$; then $\int_{B_R} |\nabla \varphi|^2 = \int_{\partial B_R} \langle \nabla_\nu \varphi, \varphi \rangle$ and for the trivial connection the only deviation from the flat model on the end is the metric, which is $O(r^{-4})$ -flat, so $D(\varphi - \varphi_\infty) = -D\varphi_\infty = O(r^{-5})$ and the decaying solution $\varphi - \varphi_\infty$ is $O(r^{-3})$ (indicial roots ≤ -3 plus a particular solution of order r^{-4}). Hence the boundary term is $O(R^{-4}) \cdot O(R^3) \rightarrow 0$ and φ is parallel. $\square$

Remark 3.3. For a non-trivial bundle, $\mathcal{K}^{+,b}(\mathcal{V})$ is in general *larger* than the space of parallel sections: the relative index theorem (crossing the indicial root 0) gives $\dim \mathcal{K}^{+,b}(\mathcal{V}) = \dim(S^+ \otimes V)^+ - \dim \mathcal{K}(\mathcal{V}) + \dim\{\varphi \in \mathcal{K}(\mathcal{V}) : \varphi = O(r^{-4})\}$. For instance, for the charge-one $SU(2)$ -instanton on $\mathbb{R}^4$ this space is three-dimensional although there are no parallel sections. This is why the proof of Theorem 4.1 below never uses “bounded positive harmonic spinors are parallel”; it uses decay along the diagonal instead.

Definition 3.4. Let $E = P \times_H G$ with the connection A induced from an H -connection A_H on P . We say that A is *G-rigid* if it is infinitesimally rigid as a G -connection, i.e. $\mathcal{H}_A^1 = \text{Ker}_{L^2} \delta_A = 0$ on $\Omega^1(X, \text{Ad } E)$. Since $\text{Ad } E = P \times_H \mathfrak{g} = \bigoplus_k \mathcal{V}_k^{\oplus m_k}$ for the irreducible H -modules V_k occurring in $\mathfrak{g}|_H$, (8) shows

$$A \text{ is } G\text{-rigid} \iff \mathcal{K}(\mathcal{V}_k) = 0 \text{ for every irreducible constituent } V_k \text{ of } \mathfrak{g}|_H.$$

This is exactly the hypothesis “ A_j infinitesimally rigid” of [GPTW, Def. 4.6 (iii), Prop. 6.2]; note that it involves all representations of H occurring in $\mathfrak{g}$, not only the adjoint representation of H (see Section 6).

Now let $\mathcal{V} = P_1 \times_{H_1} V$, $\mathcal{W} = P_2 \times_{H_2} W$ be constituents of (6), and consider the bundle $\mathcal{V} \boxtimes \mathcal{W}$ over $X \times X$ with the product connection. The twisted Dirac operator of $X \times X$ on $S(X \times X) \otimes (\mathcal{V} \boxtimes \mathcal{W}) = (S_1 \otimes \mathcal{V}) \boxtimes (S_2 \otimes \mathcal{W})$ is

$$D_B = D_1 \otimes 1 + \gamma_1 \otimes D_2, \quad D_B^2 = D_1^2 \otimes 1 + 1 \otimes D_2^2, \quad (9)$$

where $D_1 = D_{\mathcal{V}}$, $D_2 = D_{\mathcal{W}}$ and $\gamma_1 = \pm 1$ on $S_1^\pm$ (the cross terms cancel because D_1 anticommutes with γ_1). With respect to (3), a section a of $S^-(X \times X) \otimes (\mathcal{V} \boxtimes \mathcal{W})$ has components $a = (a_{-+}, a_{+-})$ and $D_B a = 0$ reads

$$D_1^- a_{-+} + D_2^- a_{+-} = 0 \quad (\text{in } S_1^+ \otimes S_2^+), \quad D_1^+ a_{+-} - D_2^+ a_{-+} = 0 \quad (\text{in } S_1^- \otimes S_2^-); \quad (10)$$

a section $b = (b_{++}, b_{--})$ of $S^+(X \times X) \otimes (\mathcal{V} \boxtimes \mathcal{W})$ satisfies $D_B b = 0$ iff

$$D_1^+ b_{++} - D_2^- b_{--} = 0 \quad (\text{in } S_1^- \otimes S_2^+), \quad D_2^+ b_{++} + D_1^- b_{--} = 0 \quad (\text{in } S_1^+ \otimes S_2^-). \quad (11)$$

(Here $D_1^\pm$ acts on the $S_1 \otimes \mathcal{V}$ factor, i.e. in the variable x , and $D_2^\pm$ in the variable y ; operators in different variables commute.)

Lemma 3.5. *Under the identifications (4) (twisted by $\text{Ad } Q$) the operator $L_B = (d_B^*, d_B^7)$, $\Omega^1(\text{Ad } Q) \rightarrow \Omega^0(\text{Ad } Q) \oplus \Omega_7^2(\text{Ad } Q)$, corresponds to $D_B: \Gamma(S^- \otimes \text{Ad } Q) \rightarrow \Gamma(S^+ \otimes \text{Ad } Q)$ up to a fixed non-zero constant on the Λ_7^2 -component, and L_B^* corresponds to $D_B: \Gamma(S^+ \otimes \text{Ad } Q) \rightarrow \Gamma(S^- \otimes \text{Ad } Q)$. In particular $\text{Ker } L_B \cong \text{Ker}(D_B|_{S^- \otimes \text{Ad } Q})$ and $\text{Ker } L_B^* \cong \text{Ker}(D_B|_{S^+ \otimes \text{Ad } Q})$ on every weighted Hölder space, and both operators preserve the decomposition (6). Moreover, with $\tilde{\delta}_{A_i} = (d_{A_i}^*, \sqrt{2}d_{A_i}^+)$ and the identifications $p_1^* \Lambda^1 \leftrightarrow S_1^- \otimes S_2^+$, $p_2^* \Lambda^1 \leftrightarrow S_1^+ \otimes S_2^-$,*

$$\tilde{L}_B^* \tilde{L}_B = \begin{pmatrix} \tilde{\delta}_{A_1}^* \tilde{\delta}_{A_1} \otimes 1 + 1 \otimes \nabla_{A_2}^* \nabla_{A_2} & 0 \\ 0 & \nabla_{A_1}^* \nabla_{A_1} \otimes 1 + 1 \otimes \tilde{\delta}_{A_2}^* \tilde{\delta}_{A_2} \end{pmatrix} \text{ on } (p_1^* \Lambda^1 \oplus p_2^* \Lambda^1) \otimes \text{Ad } Q, \quad (12)$$

which is the analogue of [GPTW, (5.1)]; and $\tilde{L}_B \tilde{L}_B^* = \nabla_B^* \nabla_B$ on $(\Lambda^0 \oplus \langle \omega_a - \mu_a \rangle) \otimes \text{Ad } Q$, while on $\Lambda_{7,\text{mix}}^2 \otimes \text{Ad } Q$ it equals $D_1^+ D_1^- \otimes 1 + 1 \otimes D_2^+ D_2^-$ (curvature terms in both variables).

Proof. For a parallel spinor ψ one computes $D(v \cdot \psi) = \sum_i e_i \cdot (\nabla_{e_i} v) \cdot \psi = (d^* v) \psi + (dv) \cdot \psi = (d^* v) \psi + (\pi_7^2 dv) \cdot \psi$ because $\Lambda_{21}^2 \cdot \psi = 0$; twisting by $\text{Ad } Q$ changes nothing, and $|\eta \cdot \psi| = c|\eta|$ for $\eta \in \Lambda_7^2$ since Λ_7^2 is irreducible. This is the standard identification used in [GPTW, proof of Prop. 5.8] (via Lewis) and in [Walpuski, *Spin(7)-instantons, Cayley submanifolds and Fueter sections*]. Rescaling the Λ_7^2 -component by a constant does not change kernels (for L^* , $(\xi, \eta) \in \text{Ker } L^*$ iff $(\xi, c\eta) \in \text{Ker } \tilde{L}^*$). The formulas for $\tilde{L}_B^* \tilde{L}_B$ and $\tilde{L}_B \tilde{L}_B^*$ are (9) together with (7) on each factor. $\square$

4 The analogue of Section 5.2

The weighted Hölder norms are those of [GPTW, Def. 5.7] (same weight $w = 1 + \min\{r_1, r_2\}$); the rescaling map $s_{j,\beta,t}$ of [GPTW, (5.10)] and Proposition 5.10 carry over verbatim, with $\mathfrak{g}$ -valued forms replaced by $\text{Ad } Q$ -valued forms.

Theorem 4.1 (analogue of [GPTW, Prop. 5.8]). *Let (Q, B) be the product connection (5) of two finite-energy ASD instantons A_1, A_2 over X which are G -rigid (Definition 3.4). Then for every $\beta < 0$*

$$\text{Ker}(L_B: C_\beta^{1,\alpha} \rightarrow C_{\beta-1}^{0,\alpha}) = 0, \quad \text{Ker}(L_B^*: C_\beta^{1,\alpha} \rightarrow C_{\beta-1}^{0,\alpha}) = 0.$$

More precisely, for $\beta \in [-3, 0)$ and without any rigidity assumption,

$$\text{Ker } L_B^* \cap C_\beta^{1,\alpha} \cong \bigoplus_k \left(\mathcal{K}^b(\mathcal{V}_k) \otimes \mathcal{K}(\mathcal{W}_k) + \mathcal{K}(\mathcal{V}_k) \otimes \mathcal{K}^b(\mathcal{W}_k) \right) \subset \Gamma((S_1^- \otimes \mathcal{V}_k) \boxtimes (S_2^- \otimes \mathcal{W}_k)), \quad (13)$$

where the right-hand side is the space of “bi-harmonic” spinors d with $D_1^- d = 0 = D_2^- d$ in the two variables; and if $\mathcal{W}_k = \mathbb{R}$ (resp. $\mathcal{V}_k = \mathbb{R}$) is a trivial constituent then the contribution of $\mathcal{V}_k \boxtimes \mathcal{W}_k$ to $\text{Ker } L_B$ is $p_1^*(\text{Ker}_{L^2} \delta_{A_1}|_{\mathcal{V}_k})$ (resp. $p_2^*(\cdots)$), exactly as in [GPTW, Lemma 5.5].

Proposition 4.2 (analogue of [GPTW, Prop. 5.9]). *Let θ_{g_1, g_2} be the flat orbifold G -connection on $(\mathbb{C}^2/\{\pm 1\}) \times (\mathbb{C}^2/\{\pm 1\})$ with holonomy (g_1, g_2) (the blow-down limit of B). Then $\text{Ker } L_\theta = 0$ and $\text{Ker } L_\theta^* = 0$ on $C_\beta^{1,\alpha} \cap C_\beta^0$ for every $\beta < 0$.*

Proof. Pull back to $\mathbb{C}^2 \times \mathbb{C}^2$, where the bundle is trivial and the connection is the trivial one; sections downstairs are $\mathbb{Z}_2 \times \mathbb{Z}_2$ -equivariant sections upstairs. The proof of [GPTW, Prop. 5.9] (Lemma 5.4 applied in each factor, then the decay) goes through word for word. $\square$

The proof of Theorem 4.1 uses [GPTW, Prop. 5.8] in the following form.

Lemma 4.3 (blow-down argument). *Let $\mathbb{E} \rightarrow X \times X$ be one of the bundles $S^\pm(X \times X) \otimes (\mathcal{V} \boxtimes \mathcal{W})$ (or any Hermitian bundle with a connection which is a tensor product of pull-backs of ASD instantons) and let c be a smooth section with*

$$\nabla_B^* \nabla_B c = 0 \quad \text{and} \quad |c| \leq C w^{\beta'} \quad \text{for some } \beta' < 0.$$

Then $c = 0$.

Proof. This is the proof of [GPTW, Prop. 5.8] with three changes. (1) Subharmonicity: $\Delta|c|^2 = 2\langle \nabla^* \nabla c, c \rangle - 2|\nabla c|^2 = -2|\nabla c|^2 \leq 0$ holds for every metric connection, so [GPTW, (5.4)] and the mean value inequality [GPTW, (5.5)] (which only uses that $X \times X$ is Ricci-flat) are available; the bounds [GPTW, (5.6)–(5.8)] follow as there. (2) Limit operator: under the rescaling ϕ_ϵ the connection $B_\epsilon = (\phi_\epsilon)_* B$ converges, in C^∞ on compact subsets of the complement of the strata $\{0\} \times \mathbb{C}^2 / \{\pm 1\} \cup \mathbb{C}^2 / \{\pm 1\} \times \{0\}$, to the flat orbifold connection θ_{g_1, g_2} of Proposition 4.2 (the region $\{r_1 \geq r_0\}$ in the g_ϵ -picture is $\{r_1 \geq r_0 \lambda_\epsilon^{-1}\}$ in the g_1 -picture, where $A_1 - \theta_1 = O(r_1^{-3})$; rescaling a 1-form by $\lambda_\epsilon \rightarrow 0$ multiplies its norm by λ_ϵ^{-1} , so $|B_\epsilon - \theta|_{g_\epsilon} = O(\lambda_\epsilon^2)$, and similarly for derivatives). Hence the Schauder estimates on compact sets away from the strata have constants independent of ϵ , and the limit c^* is a bounded section of the flat orbifold bundle with $\nabla_{\theta}^* \nabla_{\theta} c^* = 0$ away from the strata. (3) Conclusion: pulled back to $\mathbb{C}^2 \times \mathbb{C}^2$ (minus the codimension-four strata, where it is bounded) c^* is a bounded vector-valued harmonic function, hence extends harmonically across the strata and is constant by Liouville; being $\mathbb{Z}_2 \times \mathbb{Z}_2$ -equivariant with $|c^*| \leq C w^{\beta'}$ it vanishes. This contradicts the normalisation [GPTW, (5.9)] exactly as in [GPTW]. $\square$

Proof of Theorem 4.1: the cokernel. By Lemma 3.5 we must show that a section $b = (b_{++}, b_{--})$ of $S^+(X \times X) \otimes (\mathcal{V} \boxtimes \mathcal{W})$ with $D_B b = 0$ and $b \in C_\beta^{1, \alpha}$ vanishes (for one constituent $\mathcal{V} \boxtimes \mathcal{W}$ at a time). By (9), $D_B^2 b = 0$ and D_B^2 preserves the splitting (3), so

$$0 = D_B^2 b_{++} = (D_1^- D_1^+ \otimes 1 + 1 \otimes D_2^- D_2^+) b_{++} = \nabla_B^* \nabla_B b_{++}$$

by (7). Lemma 4.3 gives $b_{++} = 0$. Then (11) reduces to $D_1^- b_{--} = 0$ and $D_2^- b_{--} = 0$: $d := b_{--}$ is a section of $(S_1^- \otimes \mathcal{V}) \boxtimes (S_2^- \otimes \mathcal{W})$ which is harmonic in each variable separately, with $|d| \leq C w^\beta$.

Step A (tensor decomposition). For fixed y , $x \mapsto d(x, y)$ is a bounded solution of D_1^- , hence lies in the finite-dimensional space $\mathcal{K}^b(\mathcal{V}) \otimes (S_2^- \otimes \mathcal{W})_y$ (Lemma 3.2 (i)). Choose a basis $\varphi_1, \dots, \varphi_N$ of $\mathcal{K}^b(\mathcal{V})$ and compactly supported test sections $\chi_1, \dots, \chi_N$ with $\int_X \langle \chi_i, \varphi_j \rangle = \delta_{ij}$ (possible since $\mathcal{K}^b(\mathcal{V})$ is finite dimensional and injects into distributions). Then

$$d(x, y) = \sum_{i=1}^N \varphi_i(x) \otimes \sigma_i(y), \quad \sigma_i(y) = \int_X \langle \chi_i(x), d(x, y) \rangle dx,$$

and the σ_i are smooth, bounded sections of $S_2^- \otimes \mathcal{W}$ with $D_2^- \sigma_i = 0$ (differentiate under the integral). Hence $d \in \mathcal{K}^b(\mathcal{V}) \otimes \mathcal{K}^b(\mathcal{W})$ (algebraic tensor product).

Step B (decay along the diagonal). Order the basis so that $\varphi_1, \dots, \varphi_m$ span $\mathcal{K}(\mathcal{V})$ and $(\varphi_i)_\infty$, $i > m$, are linearly independent (Lemma 3.2 (i)). Letting $|x| \rightarrow \infty$ with y fixed,

$$\ell(y) := \lim_{|x| \rightarrow \infty} d(x, y) = \sum_{i > m} (\varphi_i)_\infty \otimes \sigma_i(y),$$

$$|\ell(y)| \leq \limsup_{|x| \rightarrow \infty} C(1 + \min\{r(x), r(y)\})^\beta = C(1 + r(y))^\beta \xrightarrow{|y| \rightarrow \infty} 0.$$

So ℓ is a decaying element of $S^-(\mathbb{R}^4) \otimes V^- \otimes \mathcal{K}^b(\mathcal{W})$, i.e. $\ell \in S^-(\mathbb{R}^4) \otimes V^- \otimes \mathcal{K}(\mathcal{W})$ by Lemma 3.2 (i); by linear independence of the $(\varphi_i)_\infty$ this means $\sigma_i \in \mathcal{K}(\mathcal{W})$ for $i > m$. Therefore

$$d \in \mathcal{K}(\mathcal{V}) \otimes \mathcal{K}^b(\mathcal{W}) + \mathcal{K}^b(\mathcal{V}) \otimes \mathcal{K}(\mathcal{W}).$$

Conversely every element of the right-hand side is smooth, satisfies $D_1^- d = D_2^- d = 0$ hence $D_B d = 0$ (with $b_{++} = 0$), and lies in $C_\beta^{1,\alpha}$ for $\beta \geq -3$: e.g. for $d = \varphi \otimes \sigma$ with φ bounded and $\sigma = O(r_2^{-3})$ one has $|d| \leq C(1+r_2)^{-3} \leq C(1+\min\{r_1, r_2\})^{-3} \leq Cw^\beta$, and similarly for derivatives. This proves (13). If A_1 and A_2 are G -rigid then $\mathcal{K}(\mathcal{V}_k) = 0 = \mathcal{K}(\mathcal{W}_k)$ for all k and the right-hand side vanishes. $\square$

Proof of Theorem 4.1: the kernel. Let $a = (a_{-+}, a_{+-})$ be a section of $S^-(X \times X) \otimes (\mathcal{V} \boxtimes \mathcal{W})$, $a \in C_\beta^{1,\alpha}$, $D_B a = 0$, so that (10) holds and, applying D_B once more,

$$(D_1^+ D_1^- \otimes 1 + 1 \otimes \nabla_2^* \nabla_2) a_{-+} = 0, \quad (\nabla_1^* \nabla_1 \otimes 1 + 1 \otimes D_2^+ D_2^-) a_{+-} = 0. \quad (14)$$

Step 1. Put $c := D_1^- a_{-+} = -D_2^- a_{+-}$, a section of $(S_1^+ \otimes \mathcal{V}) \boxtimes (S_2^+ \otimes \mathcal{W})$ with $|c| \leq Cw^{\beta-1}$ (one derivative of a). Using (7) on each factor, the commutation of operators acting in different variables, and (14),

$$\begin{aligned} \nabla_B^* \nabla_B c &= D_1^- D_1^+ c + D_2^- D_2^+ c = D_1^- (D_1^+ D_1^- a_{-+}) - D_2^- (D_2^+ D_2^- a_{+-}) \\ &= -D_1^- (\nabla_2^* \nabla_2 a_{-+}) + D_2^- (\nabla_1^* \nabla_1 a_{+-}) = -\nabla_2^* \nabla_2 c - \nabla_1^* \nabla_1 c = -\nabla_B^* \nabla_B c. \end{aligned}$$

Hence $\nabla_B^* \nabla_B c = 0$ and Lemma 4.3 (with $\beta' = \beta - 1$) gives $c = 0$, i.e.

$$D_1^- a_{-+} = 0, \quad D_2^- a_{+-} = 0. \quad (15)$$

Step 2. Put $d := D_2^+ a_{-+} = D_1^+ a_{+-}$ (by (10)), a section of $(S_1^- \otimes \mathcal{V}) \boxtimes (S_2^- \otimes \mathcal{W})$ with $|d| \leq Cw^{\beta-1}$. Then $D_1^- d = D_1^- D_2^+ a_{-+} = D_2^+ D_1^- a_{-+} = 0$ and $D_2^- d = D_2^- D_1^+ a_{+-} = D_1^+ D_2^- a_{+-} = 0$ by (15). Steps A and B of the previous proof apply to d (with $\beta - 1$ in place of β) and give $d \in \mathcal{K}(\mathcal{V}) \otimes \mathcal{K}^b(\mathcal{W}) + \mathcal{K}^b(\mathcal{V}) \otimes \mathcal{K}(\mathcal{W}) = 0$ by G -rigidity. Hence

$$D_2^+ a_{-+} = 0, \quad D_1^+ a_{+-} = 0. \quad (16)$$

Step 3. By (15), for every y the section $a_{-+}(\cdot, y)$ lies in $\mathcal{K}^b(\mathcal{V}) \otimes (S_2^+ \otimes \mathcal{W})_y$; by Step A, $a_{-+} = \sum_i \varphi_i \otimes \sigma_i$ with φ_i a basis of $\mathcal{K}^b(\mathcal{V})$ as before and σ_i bounded with $D_2^+ \sigma_i = 0$ by (16), i.e. $\sigma_i \in \mathcal{K}^{+,b}(\mathcal{W})$. Letting $|x| \rightarrow \infty$ at fixed y as in Step B, $\ell(y) = \sum_{i>m} (\varphi_i)_\infty \otimes \sigma_i(y)$ is a solution of D_2^+ which decays like $(1+r(y))^\beta$, hence vanishes by Lemma 3.2 (ii); by linear independence $\sigma_i = 0$ for $i > m$, so $a_{-+} \in \mathcal{K}(\mathcal{V}) \otimes \mathcal{K}^{+,b}(\mathcal{W}) = 0$ by G -rigidity. The same argument with the factors exchanged gives $a_{+-} \in \mathcal{K}^{+,b}(\mathcal{V}) \otimes \mathcal{K}(\mathcal{W}) = 0$. Hence $a = 0$.

For the last assertion of the theorem (trivial constituent $\mathcal{W} = \mathbb{R}$, no rigidity assumed): Step 1 still gives (15); Step 2 gives $d \in \mathcal{K}(\mathcal{V}) \otimes \mathcal{K}^b(\mathbb{R}) + \mathcal{K}^b(\mathcal{V}) \otimes \mathcal{K}(\mathbb{R}) = 0$ by Lemma 3.2 (iii); Step 3 gives $a_{-+} \in \mathcal{K}(\mathcal{V}) \otimes \mathcal{K}^{+,b}(\mathbb{R}) = \mathcal{K}(\mathcal{V}) \otimes S_2^+$, i.e. a_{-+} is the pull-back of an element of $S^+ \otimes \mathcal{K}(\mathcal{V}) \cong \text{Ker}_{L^2} \delta_{A_1}|_{\mathcal{V}}$ (see (8)), and $a_{+-} \in \mathcal{K}^{+,b}(\mathcal{V}) \otimes \mathcal{K}(\mathbb{R}) = 0$. $\square$

Remark 4.4. (a) The proof shows where the hypotheses enter: no hypothesis is needed for $b_{++} = 0$ and for $c = 0$ (these are the parts of L_B, L_B^* governed by the pure rough Laplacian, exactly as for the trivial connection in [GPTW]); G -rigidity of *both* instantons is used to kill the “bi-harmonic” spinors d and the decaying-in-one-variable spinors of Step 3. For $\text{Ker } L_B^*$ the hypothesis can be weakened to: for every constituent, $\mathcal{K}(\mathcal{V}_k) \otimes \mathcal{K}^b(\mathcal{W}_k) = 0 = \mathcal{K}^b(\mathcal{V}_k) \otimes \mathcal{K}(\mathcal{W}_k)$. In particular, for a summand $\mathcal{V} \boxtimes \mathbb{R}$ (e.g. $p_1^* \text{Ad } E^\alpha$) the cokernel vanishes automatically, since $\mathcal{K}^b(\mathbb{R}) = 0$.

(b) When $\mathcal{K}(\mathcal{V}) \otimes \mathcal{K}(\mathcal{W}) \neq 0$ for a doubly twisted constituent, (13) shows that the product connection is *obstructed* on $X \times X$: e.g. for $[L^{t_1}] \boxtimes [L^{t_2}]$ with $|t_1|, |t_2| \geq 2$ (Section 6) the obstruction space contains $\mathcal{K}(L^{t_1}) \otimes \mathcal{K}(L^{t_2})$, of dimension $\lfloor t_1^2/4 \rfloor \lfloor t_2^2/4 \rfloor$ (these are L^2 sections, decaying like $r_1^{-3} r_2^{-3}$). It would be interesting to know whether such obstructed products can be deformed to non-product $\text{Spin}(7)$ -instantons on $X \times X$ with the same asymptotics.

(c) The range $\beta \in [-3, 0)$ in (13) is only needed for the converse inclusion (elements decaying like r_i^{-3} in one variable are in C_β^0 iff $\beta \geq -3$); the vanishing statements hold for all $\beta < 0$, as in [GPTW, Prop. 5.8].

(d) The $\mathbb{R}^4 \times X$ model [GPTW, Lemma 5.5] is recovered by the same proof with X_1 replaced by $\mathbb{R}^4$: then $\mathcal{K}^b(\mathbb{R}^4) = \text{constant spinors}$, which is why on $\mathbb{R}^4 \times X$ both $\text{Ker } L$ and $\text{Ker } L^*$ contain the pull-backs of $\text{Ker}_{L^2} \delta_A$, whereas on $X \times X$ the second Eguchi–Hanson factor has no bounded negative harmonic spinors (Lemma 3.2 (iii)).

5 Consequences for Section 6 and the gluing theorem

With Theorem 4.1 and Propositions 4.2, 2.4 in hand, the proofs of [GPTW, Props. 6.2, 6.3] go through with the following changes, and [GPTW, Theorem 6.5] holds for extended compatible gluing data.

- (a) *Hypotheses.* Add: A^α, A^β are G -rigid (Definition 3.4); θ irreducible and unobstructed as before.
- (b) *Case 2.A* (accumulation on an $\mathbb{R}^4 \times X$ piece) now also occurs for $j \in \{1, \dots, 8\}$ with the non-flat connection $p_1^* A^\alpha$ (twisted by the flat holonomies $\varrho(\beta), \varrho(\tau_i)$ in the $T^4/\{\pm 1\}$ -direction, which disappear in the rescaled limit on $\mathbb{R}^4 \times X$). The argument of [GPTW] (Lemma 5.5 for $\text{Ker } L$; (5.1), Prop. A.6 (2) and rigidity for $\text{Ker } L^*$) applies unchanged.
- (c) *Case 2.B* (accumulation on an $X \times X$ piece with both coordinates bounded): replace [GPTW, Prop. 5.8] by Theorem 4.1.
- (d) *Sequences in an $X \times X$ chart with exactly one coordinate escaping to infinity.* Such sequences satisfy $\lim r_{t_i}(x_i)/t_i < \infty$ but have no convergent subsequence in $X \times X$ (this case is not separated out in [GPTW, Case 2.B], where a convergent subsequence is asserted; it is harmless there because such points also lie in a neighbouring type (ii) chart). They are treated by Case 2.A of the neighbouring type (ii) chart, after translating in the $\mathbb{R}^4$ -direction.
- (e) *Case 3.B* (neck of an $X \times X$ piece, both coordinates escaping): replace [GPTW, Prop. 5.9] by Proposition 4.2.
- (f) *Case 3.A* and *Case 1* are unchanged (in Case 1 the limit flat connection has the monodromies g_α, g_β around $S_1, \dots, S_8$, which is built into E_0).

Theorem 5.1. *Given extended compatible gluing data (Definition 2.3) with θ irreducible and unobstructed and all instantons A_j ($j = 73, \dots, 76$), A^α, A^β G -rigid, there exist $c > 0$ and, for all sufficiently small t , a $\text{Spin}(7)$ -instanton $\tilde{A}_t = A_t + a_t$ on $E_t \rightarrow M_t$ with $\|a_t\|_{C_{-1;t}^{1,\alpha}} \leq ct^{3/10}$. Near the type (iii) points $\tilde{A}_t$ is $C_{-1;t}^{1,\alpha}$ -close to the (rescaled) product connection B , which is not flat.*

6 Which instantons on X are G -rigid? The $U(1)$ case

Theorem 4.1 (like [GPTW, Prop. 6.2]) needs rigidity in *every* representation of the structure group occurring in $\mathfrak{g}$. For the $U(1)$ -instantons used in [GPTW, §7] this can be decided completely.

Let (L, A) be the $U(1)$ -instanton of [GPTW, Prop. A.7], normalised so that $c_1(L) = e^*$ is the generator of $H^2(X; \mathbb{Z}) \cong \mathbb{Z}$ with $\langle e^*, [E] \rangle = 1$, and let L^t carry the induced instanton (curvature

tF_A ; holonomy $(-1)^t$ at infinity). Since $\Lambda^1 \otimes [L^t] \otimes \mathbb{C} = S^+ \otimes (S^- \otimes L^t) \oplus S^+ \otimes (S^- \otimes L^{-t})$, by (8)

$$\dim_{\mathbb{R}} \text{Ker}_{L^2}(\delta_A|_{\Omega^1([L^t])}) = 4 \dim_{\mathbb{C}} \text{Ker}_{L^2}(D^- : S^- \otimes L^t \rightarrow S^+ \otimes L^t) =: 4\kappa(t). \quad (17)$$

Proposition 6.1. $\kappa(t) = \lfloor t^2/4 \rfloor$. In particular $\kappa(t) = 0$ if and only if $t \in \{0, \pm 1\}$.

Proof. $\text{Ker}_{L^2} D^+ = 0$ on $S^+ \otimes L^t$ by (7), so $\kappa(t) = -\text{ind}_{L^2}(D_{L^t}^+)$. The L^2 -index on the ALE space equals the Atiyah–Patodi–Singer index of the truncation $X_R = \{r \leq R\}$ as $R \rightarrow \infty$ (the metric is $O(r^{-4})$ -flat and the connection $O(r^{-3})$ -flat, so the Chern–Weil integrands are integrable and the boundary operators converge to the round $\mathbb{R}P^3$ twisted by the flat bundle ϱ_t with holonomy $(-1)^t$):

$$\text{ind}_{L^2} D_{L^t}^+ = \int_X \hat{A}(X) \text{ch}(L^t) - \frac{1}{2}(h_t + \eta_t),$$

where $h_t = \dim \text{Ker}(D_{\mathbb{R}P^3, \varrho_t}) = 0$ because $\mathbb{R}P^3$ has positive scalar curvature. The three ingredients are:

- (a) $\int_X \hat{A}(X) = -\frac{1}{24} \int_X p_1 = \frac{1}{8}$: by the APS signature theorem $\sigma(X) = \frac{1}{3} \int_X p_1 - \eta_{\text{sign}}(\mathbb{R}P^3)$ with $\sigma(X) = -1$ (intersection form (-2) for our orientation) and $\eta_{\text{sign}}(L(2, 1)) = 0$ (the general formula $-\frac{1}{p} \sum_{k=1}^{p-1} \cot \frac{\pi k}{p} \cot \frac{\pi qk}{p}$ vanishes for $p = 2$).
- (b) $\int_X \text{ch}_2(L^t) = \frac{1}{2} \int_X c_1(L^t)^2 = -\frac{t^2}{4}$: the L^2 -harmonic representative of $c_1(L^t) = te^*$ is $-\frac{t}{2}$ times the harmonic representative of $\text{P.D.}[E] \in H_c^2(X)$ (because $\langle \text{P.D.}[E], [E] \rangle = E \cdot E = -2$), and the L^2 pairing of L^2 -harmonic forms equals the intersection pairing of compactly supported classes, so $\int_X c_1(L)^2 = \frac{1}{4}(E \cdot E) = -\frac{1}{2}$.
- (c) η_t : on the round S^3 the Dirac eigenvalues are $\pm(k + \frac{3}{2})$, $k \geq 0$, with multiplicity $(k+1)(k+2)$; the $(+)$ -eigenspace is $V_k \otimes V_{k+1}$ and the $(-)$ -eigenspace is $V_{k+1} \otimes V_k$ as $SU(2)_L \times SU(2)_R$ -modules, so the antipodal map acts on them by $(-1)^k$ resp. $(-1)^{k+1}$. For the two spin structures of $\mathbb{R}P^3$ one therefore finds

$$\eta(s) = \pm \sum_{j \geq 0} (-1)^j \left[(j + \frac{3}{2})^{2-s} - \frac{1}{4}(j + \frac{3}{2})^{-s} \right], \quad \eta(0) = \pm(\frac{3}{8} - \frac{1}{8}) = \pm \frac{1}{4},$$

using $\sum_j (-1)^j (j+a)^{-s} = 2^{-s}(\zeta(s, \frac{a}{2}) - \zeta(s, \frac{a+1}{2}))$ and $\zeta(0, q) = \frac{1}{2} - q$, $\zeta(-2, q) = -\frac{1}{3}B_3(q)$. Twisting by the non-trivial flat real line bundle on $\mathbb{R}P^3$ exchanges the two spin structures, so $\eta_t = \eta_0$ for t even and $\eta_t = -\eta_0$ for t odd. The untwisted index vanishes (X has no L^2 harmonic spinors, both chiralities being killed by $D^2 = \nabla^* \nabla$), which forces $\eta_0 = +\frac{1}{4}$.

Hence $\text{ind}_{L^2} D_{L^t}^+ = \frac{1}{8} - \frac{t^2}{4} - \frac{\eta_t}{2} = -\frac{t^2}{4}$ for even t and $-\frac{t^2-1}{4}$ for odd t , i.e. $\kappa(t) = \lfloor t^2/4 \rfloor$.

Consistency check. For the complex structure I_1 , $S^- \otimes L^t \cong \Lambda^{0,1} \otimes L^t$ and D^- is $\sqrt{2}(\bar{\partial} + \bar{\partial}^*)$, so $\kappa(t)$ is the dimension of the space of L^2 Dolbeault-harmonic $(0, 1)$ -forms with values in the holomorphic line bundle $L^t = \pi^* \mathcal{O}_{\mathbb{P}^1}(t)$ on $X = T^* \mathbb{P}^1$. Algebraically, $\pi_* \mathcal{O}_X = \bigoplus_{k \geq 0} \mathcal{O}(2k)$ gives

$$h^1(X, \pi^* \mathcal{O}(t)) = \sum_{k \geq 0} h^1(\mathbb{P}^1, \mathcal{O}(t+2k)) = \sum_{k \geq 0, t+2k \leq -2} (-t-2k-1) = \lfloor t^2/4 \rfloor \quad (t \leq -2),$$

and 0 for $t \geq -1$; these classes are supported along E and are L^2 -representable, and $\kappa(t) = \kappa(-t)$ by complex conjugation. The agreement with the index computation $(0, 0, 1, 2, 4, 6, 9, \dots$ for $|t| = 0, 1, 2, 3, 4, 5, 6$) is reassuring. $\square$

Remark 6.2 (on [GPTW, Prop. 7.8]). The proof of [GPTW, Prop. 7.8] asserts $\text{Ker } \delta_{A_j}|_{\Omega^1([L^t])} = 0$ for all $t \neq 0$, arguing that the curvature term of the Weitzenböck formula “contains a commutator, and since L^t is a line bundle this commutator is in an abelian Lie algebra, hence vanishes”. I believe this step is not correct: the commutator is taken in $\mathfrak{so}(E_j)$, not in $\mathfrak{so}(2)$. Concretely, for $E = [L] \oplus \mathbb{R}$ one has $\mathfrak{so}(E) = \mathbb{R}e_{12} \oplus \langle e_{13}, e_{23} \rangle = \mathbb{R} \oplus [L]$, $F_A = f e_{12}$, and for $a = a_1 e_{13} + a_2 e_{23} \in \Omega^1([L])$ the term $[F_A, a] = f \wedge (a_1[e_{12}, e_{13}] + a_2[e_{12}, e_{23}])$ is the rotation of the $[L]$ -summand by the curvature and does not vanish; in spinor terms it is the Clifford action of the ASD form F_A on S^- , which is non-trivial. By (17) and Proposition 6.1 the kernel on $\Omega^1([L^t])$ has real dimension $4\lfloor t^2/4 \rfloor$. With $\text{Ad } E_j = s_0 \mathbb{R} \oplus s_1 [L^t] \oplus s_2 [L^{2t}]$ ($s_1 = mk$, $s_2 = \frac{m(m-1)}{2}$, [GPTW, Lemma 7.7]) this gives, for $n \geq 3$:

$$A_j \text{ on } m[L^t] \oplus \mathbb{R}^k \text{ is infinitesimally rigid} \iff m = 0 \text{ or } (m = 1 \text{ and } |t| = 1).$$

In particular the bundles $m[L^{2\xi_j+1}] \oplus \mathbb{R}^k$ with $\xi_j \neq 0$ or $m \geq 2$, and $[L^{2\xi_j}] \oplus \mathbb{R}^{k-2}$ with $\xi_j \neq 0$, do *not* satisfy the rigidity hypothesis of [GPTW, Theorem 6.5], so the four-parameter families of [GPTW, Prop. 7.8] would reduce to the members with $\xi_j = 0$, $m \leq 1$ (i.e. $E_j = [L] \oplus \mathbb{R}^{n-2}$ when $\text{hol}_\theta(\gamma)$ has a two-dimensional (-1) -eigenspace, and the flat bundle otherwise), unless the linear theory is extended to handle the kernel/cokernel by a Kuranishi argument. Geometrically the non-rigid directions are real: for $m \geq 2$ the reducible connection on $m[L]$ deforms into irreducible instantons (the Kronheimer–Nakajima moduli spaces with holonomy $-1 \in SU(2)$ at infinity are positive dimensional), and for $|t| \geq 2$ the summands $[L^t] \subset \text{Ad } E_j$ carry the harmonic spinors of Proposition 6.1. I would suggest double-checking this against the authors’ own computation; if correct it also means that in Example 2.2 one must take $t_1 = t_2 = \pm 1$ (and no repeated $[L]$ -summands), which is what the example below does.

Proposition 6.3 (an explicit $SO(5)$ example). *Let $G = SO(5)$ and define $\varrho: \pi_1^{\text{orb}}(T^8/\Gamma) \rightarrow SO(5)$ on the generators of [GPTW, Lemma 7.2] by*

$$\begin{aligned} \varrho(\alpha) &= \text{diag}(-1, -1, 1, 1, 1), & \varrho(\beta) &= \text{diag}(1, 1, -1, -1, 1), & \varrho(\gamma) &= \varrho(\delta) = 1, \\ \varrho(\tau_1) &= \varrho(\tau_2) = \varrho(\tau_3) = \varrho(\tau_7) = 1, & \varrho(\tau_4) &= \text{diag}(1, -1, 1, 1, -1), \\ \varrho(\tau_5) &= \varrho(\tau_6) = \text{diag}(1, 1, 1, -1, -1), & \varrho(\tau_8) &= \text{diag}(1, 1, -1, 1, -1). \end{aligned}$$

Then ϱ is a representation, the flat connection θ is irreducible, infinitesimally rigid and unobstructed, and together with

$$E^\alpha = [L]_{12} \oplus \mathbb{R}^3, \quad E^\beta = \mathbb{R}^2 \oplus [L]_{34} \oplus \mathbb{R}, \quad E_j = \mathbb{R}^5 \quad (j = 73, \dots, 76),$$

carrying the Gocho–Nakajima instanton on $[L]$ and the trivial connection elsewhere, it forms extended compatible gluing data in the sense of Definition 2.3, with $H_1 = SO(2)_{12}$, $H_2 = SO(2)_{34}$ and $Q = p_1^[L] \oplus p_2^*[L] \oplus \mathbb{R}$ on the $X \times X$ pieces. All instantons involved are G -rigid, so Theorem 5.1 applies and produces $\text{Spin}(7)$ -instantons on M_t which near the 64 type (iii) points are close to the non-flat product connection B , whose adjoint bundle contains the doubly twisted summand $[L] \boxtimes [L]$.*

Proof. All images are commuting involutions, so the relations of [GPTW, Lemma 7.2] reduce to $\varrho(\tau_1\tau_2) = \varrho(\tau_3) = \varrho(\tau_5\tau_6) = \varrho(\tau_7) = \varrho(\tau_1) = 1$, which hold. The elements $\varrho(\beta), \varrho(\tau_5), \varrho(\tau_6), \varrho(\tau_8)$ act on the $[L]_{12}$ -summand of E^α by $+1$ and $\varrho(\alpha), \varrho(\tau_1), \dots, \varrho(\tau_4)$ on the $[L]_{34}$ -summand of E^β by ± 1 , hence by automorphisms of the instantons; $\text{hol}_\infty(A^\alpha) = \varrho(\alpha)$ and $\text{hol}_\infty(A^\beta) = \varrho(\beta)$. Rigidity of A^α : $\text{Ad } E^\alpha = \Lambda^2([L] \oplus \mathbb{R}^3) = \mathbb{R} \oplus 3[L] \oplus \mathbb{R}^3$, and $\kappa(\pm 1) = 0$; likewise for A^β . For θ we use [GPTW, Lemma 7.1]. Write e_{ij} for the standard basis of $\mathfrak{so}(5)$; $\text{Ad } \varrho(g)$ acts on e_{ij} by the product of the i -th and j -th diagonal entries. Irreducibility: the $\text{Ad } \varrho(\alpha), \text{Ad } \varrho(\beta)$ -invariant part of $\mathfrak{so}(5)$ is $\langle e_{12}, e_{34} \rangle$, and $\text{Ad } \varrho(\tau_4)e_{12} = -e_{12}$, $\text{Ad } \varrho(\tau_5)e_{34} = -e_{34}$. Rigidity: the α, β -invariant part of $\mathbb{R}^8 \otimes \mathfrak{so}(5)$ is $\mathbb{R}_x^4 \otimes \langle e_{15}, e_{25} \rangle \oplus \mathbb{R}_y^4 \otimes \langle e_{35}, e_{45} \rangle$, and τ_4 kills

e_{15}, e_{35}, e_{45} while τ_5 kills e_{25} . Unobstructedness: the α, β -invariant part of $\Lambda_7^2 \otimes \mathfrak{so}(5)$ is $\langle e_1, e_2, e_3 \rangle \otimes \langle e_{12}, e_{34} \rangle \oplus \langle e_4, \dots, e_7 \rangle \otimes \langle e_{13}, e_{14}, e_{23}, e_{24} \rangle$ (basis of Λ_7^2 as in [GPTW, Lemma 7.5]); e_{12}, e_{34} are killed as before, and $e_{13}, e_{14}, e_{23}, e_{24}$ are killed by $\tau_8, \tau_5, \tau_4, \tau_4$ respectively. (These statements were also verified by the script `search_reps.py`, which enumerates all diagonal $SO(n)$ -representations with $\varrho(\alpha), \varrho(\beta)$ as above satisfying the automorphism constraints: none exist for $n = 4$, and 12288 exist for $n = 5$ before identifying gauge-equivalent ones.) $\square$

Remark 6.4. (a) For $G = SO(4)$ with $E^\alpha = [L]_{12} \oplus \mathbb{R}^2$, $E^\beta = \mathbb{R}^2 \oplus [L]_{34}$ no such data exist among commuting-involution representations: the element $e_{12} \in \mathfrak{so}(4)$ is fixed by $\text{Ad } \varrho$ of $\alpha, \beta, \tau_5, \dots, \tau_8$ (automorphism constraint) and of $\tau_1, \dots, \tau_4$ (determinant), and the component $e_3 \otimes e_{12}$ of $\Lambda_7^2 \otimes \mathfrak{g}$ (with e_3 the $(-, -)$ -eigenvector of (γ, δ)) is fixed unless $\text{Ad } \varrho(\gamma)e_{12} = e_{12} = \text{Ad } \varrho(\delta)e_{12}$, which makes e_{12} a fixed vector of the whole group, i.e. θ reducible. For $n \geq 5$ the extra coordinate removes the determinant constraint on τ_4 . (b) One can of course also glue $[L] \oplus \mathbb{R}^{n-2}$ along $\text{fix}(\gamma), \text{fix}(\delta)$ in the same example by choosing $\varrho(\gamma), \varrho(\delta)$ with two-dimensional (-1) -eigenspaces; the script finds many such representations.

7 Two small remarks on the text of [GPTW]

- (1) In [GPTW, Lemma 7.5] the element $e_3 = dx_{14} + dx_{23} + dx_{58} + dx_{67}$ is not in Λ_7^2 for the form Ω_0 of §2.1 (it is an eigenvector of $\ast(\Omega_0 \wedge \cdot)$ with eigenvalue -1 up to the mixed part); the correct element is $dx_{14} + dx_{23} - dx_{58} - dx_{67}$. The sign does not affect the diagonal actions of $\alpha, \beta, \gamma, \delta$ used in Lemma 7.5 and Proposition 7.6.
- (2) In the proof of [GPTW, Prop. 6.2], Case 2.B, the assertion that $\lim r_{t_i}(x_i)/t_i < \infty$ forces $\tilde{x}_i \in X \times X$ to have a convergent subsequence is not quite right, since $r_{t_i}/t_i = \min\{r_1, r_2\}$ on the $X \times X$ pieces; sequences with one coordinate escaping must be assigned to Case 2.A of the adjacent type (ii) chart (see Section 5 (d)). This is a matter of bookkeeping only.

A Scripts

Two short Python scripts accompany these notes.

- `check_spin7.py`: builds Ω_0 and the operator $\alpha \mapsto \ast(\Omega_0 \wedge \alpha)$ on $\Lambda^2 \mathbb{R}^8$, checks the eigenvalues 3 (multiplicity 7) and -1 (multiplicity 21), verifies that the ASD forms of both $\mathbb{R}^4$ -factors lie in Λ_{21}^2 and that $\omega_a - \mu_a \in \Lambda_7^2$ (Lemma 1.1), checks the basis of [GPTW, Lemma 7.5] (finding the sign in e_3), and computes the signs by which $\alpha, \beta, \gamma, \delta$ act on $e_1, \dots, e_7$.
- `search_reps.py`: enumerates diagonal representations $\varrho: \pi_1^{\text{orb}}(T^8/\Gamma) \rightarrow SO(n)$ with $\varrho(\alpha) = \text{diag}(-1, -1, 1, \dots)$, $\varrho(\beta) = \text{diag}(1, 1, -1, -1, 1, \dots)$ and the automorphism constraints of Definition 2.3, and tests irreducibility, rigidity and unobstructedness via [GPTW, Lemma 7.1] by computing common fixed vectors on $\mathfrak{g}$, $\mathbb{R}^8 \otimes \mathfrak{g}$ and $\Lambda_7^2 \otimes \mathfrak{g}$.